circuit breaker in case of overload/short circuit. It has an advantage since no replacement is required and it can be reset on elimination of fault and switched ON again. The main function of an MCB is to switch the circuit, i.e., to open the circuit (which has been connected to it) automatically when the current passing through MCB exceeds the value for which it is set. It can be manually switched ON and OFF as similar to normal switch if necessary. MCBs are of time delay tripping devices, to which the magnitude of over current controls the operating time. This means, these get operated whenever overload exist long enough to create a danger to the circuit being protected. Therefore, MCBs doesn't respond to transient loads and motor starting currents. Generally, these are designed to operate within 2.5 milli seconds during short circuit faults and 2 seconds to 2 minutes in case of overloads (depending on the level of current). MCBs characteristics are:

- Rated current up to 100 A
- Trip characteristics are not normally adjustable
- Thermal or Thermal-Magnetic operation



MCB Type	Minimum Trip Current	Maximum Trip Current
Type B	3 Ir	5 Ir
Type C	5 Ir	10 Ir
Type D	10 Ir	20 Ir

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